ModelIt!

Teacher's Guide

Why Pandemics Go Exponential

The Math Behind How One Infection Becomes a Global Crisis

Quick Reference

Grade Level	Grade 09L2
NGSS Standard	HS-LS2-6, HS-ETS1-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-6, HS-ETS1-4

Evaluate claims, evidence, and reasoning that the complex interactions in ecosystems maintain relatively consistent numbers and types of organisms in stable conditions, but changing conditions may result in a new ecosystem. Use a computer simulation to model the impact of proposed solutions to a complex real-world problem with numerous criteria and constraints, including the analysis of system interactions and component interactions.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use the SIR compartmental model to quantitatively analyze disease transmission dynamics, calculate R_0, and predict the effects of interventions on epidemic trajectories.
Disciplinary Core Idea	**LS2.C Ecosystem Dynamics / ETS1.B Developing Solutions** — Complex interactions in populations (disease transmission) can maintain or disrupt stability. Students use computational models to evaluate intervention strategies for a complex real-world problem.
Crosscutting Concept	**Systems and System Models** — Students model populations as interconnected compartments where flows between compartments depend on the state of multiple components, demonstrating coupled system dynamics.

CAST Testing Alignment

MULTIPLE CHOICE:

During an epidemic with $R_0 = 3$, approximately what percentage of the population must be immune (through infection or vaccination) to achieve herd immunity and stop sustained transmission?

CONSTRUCTED RESPONSE:

Using the SIR model, explain why a quarantine implemented when 1% of the population is infected produces dramatically different outcomes than the same quarantine at 10% infection. Include in your explanation the role of exponential growth, the coupling between compartments, and the concept of R_0.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

This lesson uses a coupled compartment model (SIR dynamics) to explore how infectious diseases spread through populations. The model demonstrates coupled system dynamics — the flow between Susceptible, Infected, and Recovered compartments depends on the state of multiple compartments simultaneously. Students will explore exponential growth in the early epidemic phase, the natural balancing feedback as susceptible populations deplete, and how interventions (quarantine and vaccination) modify the system's behavior. The lesson connects mathematical reasoning (R_0, herd immunity threshold) with public health decision-making.

The SIR Model

The SIR model is one of the foundational tools in epidemiology. It divides a population into three compartments: Susceptible (S) — people who can be infected; Infected (I) — people who are currently infectious; and Recovered (R) — people who have recovered and are immune. People flow from S to I when infected, and from I to R when they recover. The rate of flow from S to I depends on the product of S, I, and the transmission rate — making it a coupled system where both compartments affect the flow simultaneously. This coupling is what creates exponential growth: more infected people create more opportunities for transmission.

R-naught and Exponential Growth

R_0 (R-naught) is the basic reproduction number: the average number of new infections caused by one infected person in a fully susceptible population. If $R_0 > 1$, each infected person creates more than one new infection, and the epidemic grows exponentially. If $R_0 < 1$, the epidemic dies out. For context: measles has $R_0 \approx$ 12-18, COVID-19 original strain ≈ 2.5-3, seasonal flu ≈ 1.3. The exponential nature means that early in an epidemic, the infected population doubles at regular intervals. With $R_0 = 3$, one case becomes approximately 59,000 cases in just 10 generations of transmission.

Herd Immunity and Natural Balancing Feedback

The SIR model contains a natural balancing feedback loop: as people get infected and recover, the susceptible population shrinks. With fewer susceptible people available, each infected person has fewer potential targets, and the effective reproduction number drops below 1. This is herd immunity — when enough people are immune that sustained transmission is impossible. The threshold is calculated as $1 - (1/R_0)$. For $R_0 = 3$, that's 67% immune. However, reaching herd immunity through natural infection means 67% of the population gets sick — with potentially catastrophic consequences.

Interventions: Changing the System

Public health interventions work by modifying specific parameters of the SIR model. Quarantine reduces the Contact Rate, directly lowering R_0. Vaccination moves people from Susceptible to Recovered without going through Infected, reducing the susceptible pool and lowering the effective R_0. The critical insight is that intervention TIMING dramatically affects outcomes. In an exponential system, acting one week earlier can prevent more infections than acting one month later with greater intensity. This is because exponential growth means the infected population during that week of delay may double or triple, creating a vastly larger base for subsequent transmission.

Beyond Basic SIR

Real epidemics are more complex than the basic SIR model. Asymptomatic transmission means some infected people spread disease without knowing they're sick. Waning immunity can turn the model into SIRS (where Recovered people become Susceptible again). Heterogeneous mixing means some people have many more contacts than others. Hospital capacity introduces a threshold effect: below capacity, mortality is manageable; above capacity, mortality spikes because patients can't

receive care. Despite these complexities, the basic SIR model captures the essential dynamics that make pandemics so dangerous and interventions so time-sensitive.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify seven components across three types: three internal compartment variables (Susceptible, Infected, Recovered), one internal rate (Transmission Rate), and three external controls (Contact Rate, Vaccination Rate, Quarantine Effectiveness). Students recognize that the three compartments are coupled — changes in one directly affect flows to the others.

E **Establish Relationships**

Students map the coupled flows: Susceptible → Infected (driven by contact between S and I) and Infected → Recovered (driven by recovery time). They identify the reinforcing loop in early epidemic (more I → more new infections) and the natural balancing loop (more R → fewer S → fewer new infections). They determine how each external control modifies these flows.

V **Visualize & Model**

Students build a coupled compartment model showing how Contact Rate and Transmission Rate drive the flow from Susceptible to Infected, how recovery drives the flow from Infected to Recovered, and how Vaccination Rate and Quarantine Effectiveness modify these dynamics. The model must demonstrate exponential growth, natural peak, and decline.

E **Evaluate Outcomes**

Students run uncontrolled, early intervention, late intervention, and vaccination scenarios. They quantify the difference in total infections and peak infected population between early and late quarantine, discovering that timing is often more important than intensity in exponential systems.

R **Revise & Extend**

Students extend the model with Hospital Capacity (threshold effect on mortality), Asymptomatic Transmission Rate, or Reinfection Rate to explore how real-world complexity changes the basic SIR dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.

At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

3 min

Say: *"In December 2019, a single person in Wuhan, China had a cough. Three months later, the entire world was in lockdown. How did ONE infection become a GLOBAL pandemic?"*

Watch For

Show a visualization of COVID-19's global spread over time. Let students react. Ask: 'What mathematical pattern could explain something spreading this fast?'

Slide 2: Learning Objectives

4 min

Say: *"Today we're building the exact type of model that scientists used to predict COVID-19's spread and to design the interventions that saved millions of lives."*

Watch For

Introduce the SIR framework visually. Pre-teach R_0 with a concrete example: 'If one person infects 3 others, and each of those infects 3 more, how many are infected after 10 rounds?' Let students calculate (59,049).

Slide 3: Big Question

4 min

Say: *"The answer is exponential growth. And the terrifying thing about exponential growth is that it's almost impossible for human brains to intuit. Let me show you why."*

Watch For

Classic doubling problem: 'If a pond has lily pads that double daily and the pond is full on day 30, when was it half full?' (Day 29.) 'When was it 1% full?' (Day 23.) This illustrates why exponential problems seem manageable until they're catastrophic.

Slide 4: LEVER Framework

3 min

Say: *"We're building a coupled compartment model — three population groups connected by flows that depend on each other. This is the most sophisticated model type we've built."*

Watch For

Draw S, I, and R boxes on the board with arrows. Ask: 'What determines how fast people move from S to I? What determines how fast they move from I to R?' Build intuition for the coupled dynamics.

Slide 5: Activity 1: Components

8 min

Say: *"We have three populations that people flow between, plus external controls that modify the flows. Sort them — but think carefully about which are internal dynamics and which are human decisions."*

Watch For

Guide component sorting. Key insight: Susceptible, Infected, and Recovered are coupled internal compartments. Contact Rate, Vaccination Rate, and Quarantine Effectiveness are external because humans control them. Transmission Rate is internal (determined by pathogen biology).

Slide 6: Activity 2: Connections

10 min

Say: *"Map the flows between compartments. The critical question: why does the flow from S to I depend on BOTH the number of Susceptible AND the number of Infected people?"*

Watch For

Students map coupled flows. Emphasize: new infections require BOTH a susceptible person AND an infected person to make contact. This coupling is why the SIR model is more complex than a simple linear system. Identify where each intervention (quarantine, vaccination) modifies the system.

Slide 7: Activity 3: Simulation

12 min

Say: *"Four scenarios. Uncontrolled epidemic, early quarantine at 1%, late quarantine at 10%, and vaccination. Predict FIRST, then run. The timing comparison will surprise you."*

Watch For

Students predict outcomes before running each scenario. Key revelation: the same quarantine at 1% vs. 10% can reduce total infections by half or more. Have students calculate: if the epidemic doubles every 3 days, how many more people are infected in the 2 weeks between 1% and 10%? This quantifies the cost of delay.

Slide 8: Discoveries

5 min

Say: *"Your model just showed you what every epidemiologist tried to tell the world in March 2020: in exponential systems, acting one week early saves more lives than acting one month late with ten times the resources."*

Watch For

Connect model findings to real COVID-19 data. Compare countries that locked down early vs. late. Discuss: 'Why is it so hard for leaders to act early, when the numbers still look small?' This is the core challenge of exponential thinking.

Slide 9: STEM Challenge

12 min

Say: *"Build a tool that helps health officials make the hardest decision in public health: when to act. Your tool needs to show what happens if they act NOW versus if they wait."*

Watch For

Groups design decision support tools using model data. Key requirement: the tool must show the 'cost of waiting' clearly. Groups present and peer-evaluate on usability, accuracy, and decision-support value.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Contact Rate, Vaccination Rate, Quarantine Effectiveness
- **INTERNAL:** Susceptible Population, Infected Population, Recovered Population, Transmission Rate

Relationships

- **Contact Rate to Infected Population:** POSITIVE (+) — Higher Contact Rate means each infected person encounters more susceptible people per day, increasing the rate of new infections. This is the primary target of quarantine interventions.
- **Susceptible Population to Infected Population:** POSITIVE (+) [Coupled] — A larger Susceptible Population provides more potential targets for infection. As S decreases (through infection or vaccination), the rate of new infections naturally slows — this is the balancing feedback that eventually ends epidemics.
- **Infected Population to Susceptible Population:** NEGATIVE (-) [Coupled] — More Infected individuals deplete the Susceptible population faster, as susceptible people become infected. This coupling means the flow rate depends on BOTH compartments simultaneously.
- **Infected Population to Recovered Population:** POSITIVE (+) — More Infected individuals means more people recovering over time, increasing the Recovered population and building herd immunity.
- **Vaccination Rate to Susceptible Population:** NEGATIVE (-) — Vaccination moves people directly from Susceptible to Recovered, reducing the susceptible pool without requiring infection. This is the most efficient path to herd immunity.
- **Quarantine Effectiveness to Contact Rate:** NEGATIVE (-) — Higher Quarantine Effectiveness reduces the effective Contact Rate, lowering R_0 and slowing transmission. This intervention modifies the system's behavior without changing the pathogen itself.
- **Transmission Rate to Infected Population:** POSITIVE (+) — Higher Transmission Rate means each contact is more likely to result in infection, increasing the flow from S to I and driving higher R_0.

Key Discoveries

- Pandemics grow exponentially because every infected person creates multiple new infections — each of whom creates multiple more
- The SIR model shows that epidemics naturally peak and decline as the susceptible population is depleted, but the damage before the peak can be catastrophic
- Timing of intervention matters enormously: the same quarantine implemented at 1% vs. 10% infection can reduce total infections by 50% or more
- Vaccination creates a shortcut from Susceptible to Recovered, building immunity without the disease — reducing both the peak and total infections
- R_0 determines epidemic dynamics: above 1, the epidemic grows; below 1, it shrinks. Every intervention aims to push R_0 below 1
- The three compartments (S, I, R) are coupled — changes in one directly affect the flows to and from the others

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Pandemics grow at the same rate throughout"	Pandemics do NOT grow at a constant rate. They grow exponentially at first (when the susceptible pool is large), reach a peak, then decline as the susceptible population is depleted. The epidemic curve is bell-shaped, not linear. The model shows this clearly: the growth rate depends on the SIZE of both the Susceptible and Infected populations, which change over time.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Quarantine and vaccination do the same thing"	Quarantine and vaccination work on DIFFERENT components of the SIR model. Quarantine reduces the Contact Rate — it doesn't change who's susceptible, but it reduces how often susceptible and infected people interact. Vaccination reduces the Susceptible Population — it moves people directly to Recovered without infection. Quarantine is temporary and must be maintained; vaccination is permanent (for that individual). Both reduce R_0, but through entirely different mechanisms.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Herd immunity means nobody gets sick"	Herd immunity doesn't mean zero infections — it means sustained TRANSMISSION stops. Once enough people are immune, each infected person infects fewer than one new person on average, and chains of transmission die out. But individuals can still get infected; they just don't spark chain reactions. Herd immunity protects the POPULATION, not every individual.